William Chuang — Curriculum Vitae

Department of Mathematics
University of Arizona
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Education

University of Arizona
M.S. in Mathematics (Expected) — Spring 2025
(Graduate-level coursework completed under Ph.D. classification)

San Francisco State University
M.A. in Mathematics — Spring 2022
Thesis: *The Hausdorff Dimension of Limit Sets of Well-Distributed Schottky Groups*
Link: `https://scholarworks.calstate.edu/downloads/xd07h079g`
Advisor: Dr. Chun-Kit Lai

University of San Francisco
B.S. in Mathematics, Minor in Computer Science — Fall 2018
Major GPA: 3.88/4.00
Graduated with Honors

Research Interests

I am broadly interested in the interplay between mathematical physics, geometry, and the topology of data. My recent work focuses on:

- **Developing Quantum-Ready Transformer/Autoencoder Architectures from Geometric First Principles:** Constructed a novel framework wherein the dynamics of attention and encoding layers are governed by Möbius flows in $\mathrm{PSL}(2,\mathbb{R})$, derived directly from differential equations of the form $\frac{dz}{dt} = \kappa(z - z_-)(z - z_+)$ associated with isometries. Each layer corresponds to the exponential of a Lie algebra element in $\mathfrak{sl}(2,\mathbb{R})$, naturally forming a one-parameter flow that preserves hyperbolic geometry. This system:
  - Enables exact quantization via Schrödinger-type evolution under unitary group representations;
  - Encodes model evolution as geodesic motion in curved latent spaces, improving interpretability and structure-preserving compression;
  - Outperforms existing quantum ML proposals by unifying classical geometry and quantum operator theory from first principles;
  - Provides a new foundation for quantum LLMs where activation and attention flows evolve by analytically integrable, symmetry-conserving dynamics.

- Analyzing the intrinsic factors and topological structures underlying self-attention architectures, aiming to understand how geometry and topology inform algorithmic design.
- Developing theoretical frameworks in geometric data analysis, notably through an abstract definition of curvature for novel data representations.
- Applying hyperbolic geometry and Kleinian groups to enhance neural network performance and security—specifically:
  (1) Introducing global symmetries—recognized as a parallel to Poincaré's monodromy—into neural networks, particularly in autoencoders and transformers, by leveraging group actions on these networks. This extends monodromy beyond its classical role in Fuchsian differential equations to all intermediate solution paths in a neural network, structuring the entire optimization landscape through group-theoretic mappings.
  (2) Strengthening data encryption and decryption,
  (3) Devising new approaches for deriving neural networks that can be composed or edited mathematically based on previously trained models,
  (4) Dynamically modifying energy landscapes on a global scale to reduce computational costs during training by sequentially mapping model parameters to configurations that systematically decrease the loss function's output.
- Investigating linear combinations, cyclic groups, and orbifolds of feedforward neural networks with restricted codomains, leveraging autoencoders composed of feedforward networks and their inverses to achieve deeper model interpretability.
- Constructing a rigorous interaction theory by integrating:
  (1) The Callan–Symanzik equation and stochastically motivated measures,
  (2) Motives and Hopf algebras within a renormalization group (RG) framework,

  to advance the mathematical understanding of deep neural networks.

Original Contributions and Historical Firsts

The following research contributions represent original discoveries, each constituting a historical first in its respective domain, based on available literature and records:

(1) **Redefining Core Algebra in Neural Networks Using Hyperbolic Geometry:** This directly unlocks model orbit mobility under group actions—achieving a level of control and transformation that Marvin Minsky famously proved to be computationally intractable, even NP-hard, in the classical perceptron model (cf. *Perceptrons*, 1988, with Papert).

- *Historical First:* No one has previously published a fully geometrically consistent, hyperbolically grounded framework in deep learning that enables this level of group-based model orbit mobility.
- *Recent Extension:* This foundational structure has now been elevated to support an infinite family of mathematically consistent geometric methods—each preserving

the expressive power of conventional Euclidean architectures, including the ability to represent negative-valued parameters and outputs. These constructions provide a unified, invariant framework for building deep neural networks, transformers, autoencoders, and any future architectures constructed from conventional deep learning primitives. Each variant within this framework may be optimally suited to different use cases depending on geometric, topological, analytical, algebraic, or statistical/probabilistic considerations—opening a new paradigm for principled architecture design in intelligent systems, first-principles quantum neural networks, and post-machine-learning and post-quantum cryptographic applications.

(2) **First Quantum-Ready Transformer Architecture Built from Geometric Principles:** Developed a quantum transformer and autoencoder design where each layer corresponds to Möbius transformations generated by Lie algebra symmetries. This structure naturally evolves through Schrödinger-type dynamics and is built from first principles rather than heuristic circuit designs.

(3) **Monodromy of Neural Networks:** Introduced the idea of tracking how a deep learning model evolves globally as its parameters move in loops—extending the classical concept of monodromy to the training dynamics of modern AI models.

(4) **Topological Entropy as a Cryptographic Hardness Parameter:** Embedded the Hausdorff dimension δ into cryptographic design as a tunable constant that governs complexity, obfuscation, and resistance against both classical and quantum attacks.

(5) **Neural Encryption via the Holographic Principle:** Designed a cryptographic neural architecture grounded in the holographic principle, where data and parameters are embedded in curved latent spaces. Established a formal bulk–boundary correspondence:
   - Input tokens $\leftrightarrow$ boundary observables,
   - Transformer layers $\leftrightarrow$ bulk fields in latent space,
   - Weight dynamics $\leftrightarrow$ renormalization flow,
   - Cipher inversion $\leftrightarrow$ boundary reconstruction from bulk geometry,
   - Semantic representation $\leftrightarrow$ automorphic kernel functions.

   The encryption mechanism requires full latent structure for decryption, making partial inversion equivalent to reconstructing bulk geometry from boundary data—thus encoding strong cryptographic hardness rooted in holographic duality.

(6) **AdS/CFT-Inspired Neural Cryptosystem:** Explicitly modeled a neural architecture on the AdS/CFT correspondence. The bulk geometry was implemented using hyperbolic latent space ($\mathbb{H}^{n+1}$), with transformer layers representing bulk fields and outputs interpreted as boundary CFT observables. The system reaches a theoretical security tier informally comparable to an "S++ class" in analogy with NSA-style classification schemes, due to its reliance on unsolved quantum gravity dualities and automorphic obfuscation. Constructed a structural map:
   - Tokens $\leftrightarrow$ CFT operators on the boundary,
   - Transformer depth $\leftrightarrow$ radial AdS coordinate,
   - Parameter flow $\leftrightarrow$ RG flow in bulk field theory,
   - Model inversion $\leftrightarrow$ AdS/CFT duality inversion,

- Compression and encoding $\leftrightarrow$ minimal surfaces (Ryu–Takayanagi formula).
- *Implementation Clarification:* Importantly, this approach does not assume our physical universe is Anti-de Sitter. Instead, the AdS "bulk" is rendered as a computational geometry within the model—analogous to how physics engines simulate curved spaces in virtual environments. The duality is symbolically encoded in the model's tensor structure, with geodesic kernels and partition-function-based attention. Thus, the AdS/CFT framework is not a metaphysical claim but an engineered duality, implemented purely through transformer design and hyperbolic geometry, and fully realizable on conventional hardware.

This cryptosystem achieves maximal obfuscation by leveraging the intractability of reconstructing bulk geometry from boundary data—an encryption-hardness analog of AdS/CFT inversion.

(7) **Embedding Langlands Duality in Neural Cryptographic Design:** Proposed a GPT-style transformer architecture in which *Langlands reciprocity* is structurally encoded. This architecture connects neural network representations with dual objects in arithmetic geometry, establishing a functorial correspondence between modular forms and cryptographic keys. The framework uses Galois representations and automorphic eigenstructures to algebraically infer obfuscation parameters.
- *Historical First:* This is the first known application of Langlands duality principles to cryptographic architecture design in artificial intelligence.

(8) **First Use of Automorphic Forms in Attention Mechanisms:** Introduced automorphic kernel functions—specifically Maass–Poincaré series and Hecke eigenfunctions—into transformer attention layers. This approach integrates deep structures from harmonic analysis into machine learning, enabling modular-invariant attention propagation and enhancing both representation stability and security.
- *Historical First:* No prior work in neural network theory or practice has used automorphic forms as the basis for attention mechanisms.

(9) **Loss Landscape Control via Lorentz Transformations:** Discovered that Lorentz transformations, as isometries of Minkowski space, can be used to reshape and reparameterize neural loss landscapes. This curvature-aware method introduces a relativistic alternative to gradient descent, enabling loss tunneling, gradient teleportation, and global convergence via lightlike or timelike geodesic paths.
- *Historical First:* This is the first formulation of Lorentzian geometry as a landscape-control principle in machine learning optimization.

(10) **Langlands–Neural Correspondence as a Meta-Training Theory:** Developed a novel mathematical–computational framework connecting the Langlands program in number theory to neural network training dynamics. This theory constructs a categorical functor between automorphic representations (e.g., modular forms, Maass eigenfunctions) and Galois representations (e.g., Frobenius eigenvalues), enabling closed-form, symbolic prediction of neural weight updates and hyperparameter evolution.

- *Historical First:* First formal realization of the Langlands correspondence in a machine learning context, allowing explicit prediction of training outcomes through dual algebraic representations, bypassing conventional gradient descent.
- *Structural Innovation:* Defines a Langlands functor $\mathcal{L} : \mathcal{A} \to \mathcal{G}$ between categories of automorphic forms $\mathcal{A}$ and Galois representations $\mathcal{G}$, with functorial update rules $\theta_{t+1} = \mathcal{L}^{-1}(\rho_t(\mathrm{Frob}_p))$ replacing heuristic optimization.
- *Topological–Algebraic Integration:* This theory complements and extends prior work on monodromy-based training, where monodromy describes the topological evolution of models via looped parameter flows, and Langlands duality provides algebraic targets for global minima. Together, they form an algebraic–topological hybrid mechanism for landscape reconfiguration and global optimization.
- *Training Outcome:* Enables computation of optimal architectures, weights, and critical points using modular eigenvalue structures, Satake parameters, and inverse functorial transport over dual representation categories.

(11) **Modular Spectral Indistinguishability Assumption (MSIA) – Post-Quantum Cryptographic Primitive:** Introduced the MSIA framework, a novel cryptographic assumption built on modular spectral factorization, symbolic dynamics, and modular representation theory. The core challenge is the inversion or factorization of structured zeta functions over finite fields:

$$\zeta_p(z) = \frac{1}{\det(I - zA^{(p)})}, \quad A^{(p)} \in \mathbb{F}_p^{n \times n}.$$

The structure embeds rich combinatorial and algebraic constraints, including projective indecomposable modules and symbolic Brauer classes.

- *Security Model:* Achieves theoretical hardness analogous to NP-hard and #P-hard problems under both classical and quantum adversarial models. The framework supports trapdoor-based encryption with up to 100× parameter amplification, enabling layered obfuscation mechanisms.
- *Classification:* Under informal cryptographic hardness analogies (e.g., C, B, A, S, S+), MSIA with appropriate scaling can reach the S+ level, and with additional symbolic or hybrid obfuscation layers, may approach an S++-class cryptosystem in terms of theoretical resistance to inversion. These terms are used metaphorically to denote exceptional post-quantum intractability and are not tied to formal government classification levels.
- *Historical First:* First known use of modular trace zeta functions and symbolic module structures to construct post-quantum cryptographic hardness assumptions within a structured, spectral-theoretic framework.

Independent Mathematical Theories in Neural Architecture, Cryptography, and Mathematical Physics

The following original contributions each satisfy the formal requirements of a mathematical theory: *explicitly defined objects, internal axioms, morphisms (or dynamics), and closure* *under composition.* These frameworks serve as high-level generalizations that unify, simplify, or predict structural phenomena in deep learning, post-quantum cryptography, and quantum field theory. Each theory has been rigorously constructed to support independent module development and formal derivation, and many admit categorical formulations. They span foundational neural algebra, quantum dynamics, symbolic cryptography, and arithmetic representation theory. Several are interconnected and modular, forming a self-consistent architecture of novel post-quantum and intelligence-relevant learning systems.

(1) **GRAIL (Geometric Representation Algebra for Intelligent Learning)** — Universal Meta-Architecture for Geometry-Based Neural Computation
  - GRAIL encodes neural computation as algebraic operations over curved manifolds (e.g., hyperbolic, Lorentzian, modular), generalizing neural architectures beyond Euclidean space.
  - Implementations based on representation-theoretic linear functionals represent a finite subclass within GRAIL.
  - GRAIL more broadly admits nonlinear, non-symmetric, non-metric operations derived from automorphic kernels, quantum deformations, and symbolic-entropic dynamics.
  - These generalized operators support encryption, symbolic attention, post-quantum obfuscation, and arithmetic alignment—subsuming and extending all traditional deep learning frameworks.
  - Defines neural computation over general curved spaces through group-theoretic and categorical structures, including Lie algebra symmetries and equivariant morphisms.
  - **Training Regimes:**
    - (1) Backpropagation-compatible curved-space operators, suitable for gradient-based optimization.
    - (2) Symbolic, non-differentiable operators (e.g., Langlands layers, cryptographic kernels) trained via monodromy, Lorentzian landscape reshaping, fixed-point iteration, or categorical dynamics.
  - **Satisfies:** Generalized geometric axioms, symmetry group closure, nonlinear operator composition, and categorical consistency across transformation layers.

(2) **First-Principle Quantum Neural Dynamics (FPQND) — Theory of Quantum Neural Evolution:** Establishes a framework where quantum neural nets evolve through Schrödinger-type equations derived from first-principles (Lie group symmetries, Hamiltonian flows).
  - *Satisfies:* Axiomatized operator evolution, Lie-algebraic generators, tensor flow conservation laws, and category-consistent time dynamics.

(3) **Langlands–Neural Correspondence (LNC) — Categorical Theory of Arithmetic–Neural Duality:** A global categorical framework embedding Langlands duality into neural training dynamics. Maps automorphic and Galois representations into the evolution of model parameters, layer-wise structure, and loss landscape trajectories. Defines functorial update rules replacing conventional gradient descent using arithmetic spectral data and categorical lifting mechanisms.

- *Satisfies:* Langlands functoriality, trace axioms, Frobenius update mappings, and dual group correspondence. Applies to weight dynamics, spectral phase transitions, and structural generalization rules.
- **Derived Architecture – Quantum–Langlands Attention Integration (QLAI):** Implements LNC in attention layers using automorphic kernel activations (e.g., Maass forms) and Hecke operator-based spectral filtering. This is the first concrete realization of Langlands theory in transformer-based architectures.

(4) **Monodromic Training Dynamics (MTD) — Theory of Topological Model Shifts:** Uses monodromy theory and homotopy group actions to encode training progress via parameter loops, enabling topological descent mechanisms and nonlocal optimization.
- *Satisfies:* Loop consistency, homotopic deformation axioms, covering space dynamics, and commutative training loops.

(5) **Curved-Space Neural Encryption (CNE) — Cryptographic Duality Theory over Curved Architectures:** A general cryptographic framework built from geometric dualities between neural bulk layers and latent boundary representations. This theory extends AdS/CFT-inspired architectures to arbitrary curved spaces defined by GRAIL, enabling secure, non-invertible encodings through bulk–boundary transformations. In practice, hyperbolic models provide the first concrete implementation, but the framework supports any curved space satisfying GRAIL axioms.
- *Satisfies:* Duality-based cryptographic hardness, functorial bulk–boundary maps, geometric injectivity constraints in latent space projections.
- *Note:* Hyperbolic neural encryption is an instantiation of this theory, but the duality formalism is geometry-agnostic and generalizes across all admissible GRAIL manifolds.

(6) **Modular Spectral Indistinguishability Assumption (MSIA) — Post-Quantum Cryptographic Theory:** Introduces a new cryptographic hardness assumption based on symbolic dynamics, zeta function factorization, and modular trace algebra.
- *Satisfies:* Algebraic structure of zeta matrices, symbolic representation modularity, and entropy-consistent trapdoor structures.

(7) **Fractal Cryptographic Dynamics (FCD) — Entropic Obfuscation Theory:** Defines security primitives using fractal dimension and symbolic entropy as tunable cryptographic hardness parameters.
- *Satisfies:* Measure-theoretic invariance, chaotic symbolic encoding, dimensional tunability.

(8) **Arithmetic–Neural Quantum Field Theory (ANQFT) — Universal Theory of Loop Integrals via Neural Objects:** Proposes a method to encode all-loop-order Feynman integrals and quantum periods using neural-algebraic automorphic forms.
- *Satisfies:* Functorial mapping of Feynman integrals to neural representations, categorical decomposition of modular periods, and explicit simplification via neural moduli.

(9) **Cosmic Neural Galois Representation (CNGR) — Theory of Arithmetic Symmetries in Learnable Systems:** Defines a category of neural-motivic objects encoding

mixed motives, Galois actions, and period structures. Constructs Frobenius-compatible updates, L-function representations, and symbolic descent mechanisms that simulate the action of the cosmic Galois group within learnable architectures.

- *Satisfies:* Functorial Galois realizations, motivic descent rules, spectral L-value generation, and categorical representation of arithmetic periods within neural systems.
- *Applications:* Symbolic arithmetic learning, automorphic AI alignment, arithmetic period inference, and post-quantum cryptographic resilience via neural–motivic duality.

National Security Alignment and Clearance Readiness

The theoretical frameworks developed herein—spanning neural geometry, post-quantum cryptography, Langlands duality, monodromy-based training, and hyperbolic transformers—have been constructed with clear structural relevance to national security, cryptanalytic resilience, and post-quantum computing strategy.

A portion of this work, titled *"Hyperbolic Autoencoder and Transformer Framework"*, was submitted as part of a formal RFI Response to **DARPA-SN-25-60** and is provisionally protected under **U.S. Patent Application #63/773,441** (filed March 18, 2025). This disclosure outlines the use of:

- Monodromy-driven hyperbolic transformer layers,
- Lorentzian landscape modulation,
- Langlands correspondence for embedding post-quantum security directly into attention mechanisms.

While many theoretical components remain unpublished to preserve strategic and export-sensitive value, all research has been conducted in compliance with U.S. academic and regulatory standards, including ITAR and EAR public disclosure protocols.

Clearance Readiness: Open to future clearance-based collaboration with U.S. national security agencies (e.g., **DoD, DARPA, NSA, CIA, FBI**), subject to eligibility and review procedures. All current materials remain at the unclassified level and are suitable for academic dissemination while retaining applicability to mission-critical domains.

Other Independent Projects

University of Arizona

- **Independent Study: Real and Complex Analysis, and Applications of Hyperbolic Geometry**
  With Prof. David Glickenstein, Fall 2023

University of Arizona

- **RTG Project: Scaling Factors of Self-Attention Weights in Transformers**
  With Prof. Ning Hao, Fall 2023

San Francisco State University

- **Computation of the Hausdorff Dimension of Limit Sets of Schottky Groups**
  With Dr. Chun-Kit Lai, June 2021 – May 2022

San Francisco State University

- **Independent Study: Prime Geodesic Theorem and Limit Sets of Schottky Groups**
  January 2021 – May 2021
  Wrote a summary of the modern proof with an emphasis on growth rates based on the Hausdorff dimension of the associated limit set.
  Advisor: Dr. Chun-Kit Lai

San Francisco State University

- **Topology Project: A Study on Fundamental Groups**
  September 2020 – December 2020
  Advisor: Dr. Emily Clader

San Francisco State University

- **Independent Study: Hom-Polytopes**
  September 2019 – December 2019
- **Combinatorics Project: Simplicial Complexes**
  January 2019 – May 2019
  Advisor: Dr. Joseph Gubeladze

University of San Francisco

- **Independent Study: Prime Number Theorem**
  January 2018 – May 2018
  Advisor: Dr. Paul Zeitz

Pennsylvania State University–University Park

- **Functional Analysis Project: Hardy's Proof of Uniform Distribution**
  January 2018 – May 2018
- **Independent Study: Reading "Lecture Notes on Functional Analysis: With Applications to Linear Partial Differential Equations"**
  January 2018 – May 2018
  Advisors: Dr. Sergei Tabachnikov and Dr. Moisey Guysinsky

Pennsylvania State University–University Park

- **Topology Project: Solving the $(9, 8, 4, 3, 7)$-Linkage Problem**
  January 2018 – May 2018
- **Topology Final Project: Conway's Basic Theorem**
  September 2017 – December 2017
  Advisor: Dr. Sergei Tabachnikov

University of San Francisco

- **Capstone Project: Graph Theory for an Inverted-Index-Based Search Engine**
  January 2018 – May 2018
  Advisor: Dr. Chris Bryan

University of San Francisco

- **Capstone Project: Applying the Method of Steepest Descent and Cauchy Contour Integrals to the Fisher Exact Test**
  January 2018 – May 2018
  Advisor: Dr. Xuemei Chen

University of San Francisco

- **Research Assistant**
  August 2016 – May 2017
  Assisted with lecture notes for MSAN 504 (Review of Probability and Statistics).
  Advisor: Dr. Jeff Hamrick

University of San Francisco

- **Capstone Project: Implementing Dijkstra's Algorithm Applications**
  Spring 2016
- **Summer Research Project: Therapeutic Video Games for Patients with Disabilities**
  June 2016 – September 2016
- **Interpreting Deep Neural Networks**
  Fall 2016
  Explored causal structures within deep networks to map them onto symbolic graphs, and investigated methods to initialize models from human-written code.
  Read causal inference research by Prof. David Galles and Judea Pearl.
  Advisor: Dr. David Galles

Pre-Baccalaureate Independent Projects

National Taiwan University

- **Research Student at LeCosPA**
  September 2011 – May 2013
  Presented various topics in weekly meetings and seminars, including:
  - Bremsstrahlung and Cherenkov radiation
  - Topological quantum field theory and 2+1D quantum gravity via Chern-Simons terms
  - Cosmological constant, vacuum structure, and vacuum energy
  - Radiation from moving mirrors and black holes (Schwinger mechanism, Casimir effect, Hawking/Unruh effects)
  - Potential carbon-free energy sources via low-energy nuclear reactions
  - Metamaterials and analog models of gravity
  - Instability of Anti-de Sitter space
  - Induced gravity, Coleman-Weinberg–Witten theorem on Lorentz violation, AdS/CFT correspondence
  - Quantum information, holographic turbulence, AdS/CMT, sonoluminescence

  - Holographic renormalization group flow and Ricci flow
  - Background-independent spin foam models and Regge calculus

  Advisor: Dr. Pisin Chen

National Taiwan University

- **Kontsevich–Soibelmann Wall-Crossing Formula for Mathematical QFT**
  January 2012 – May 2012
- **Conformal Bootstrap Methods for the 3D Ising Model**
  2011
  Advisor: Dr. Heng-Yu Chen

National Taiwan University

- **A Study on the Lee–Yang Theorem and Riemann Zeta Function in Statistical Mechanics**
  January 2012 – May 2012
  Advisor: Dr. Ning-Ning Pang

National Taiwan University

- **Dark Energy Problem via Modified Gravity**
  September 2010 – May 2011
  Focused on the equivalence of Einstein frames and conformal mappings in scalar-tensor theory.
  Advisor: Dr. Je-An Gu

Work Experience (Teaching & Research)

University of Arizona

- Graduate Teaching Assistant, MATH 112 (College Algebra), Section 33 – Fall 2022
- Graduate Teaching Assistant, MATH 112 (College Algebra), Sections 12 & 18 – Spring 2023
- Graduate Teaching Assistant, MATH 112 (College Algebra), Section 13 – Fall 2023
- Grader, MATH 112 (College Algebra), Sections 9, 13 & 20 – Fall 2023
- Tutor, MATH 129 (Calculus II) – Fall 2023
- Grader, MATH 129 (Calculus II) Final Exam – Fall 2023
- Grader, MATH 122B/125 (Calculus I) Common Final Exam – Fall 2023
- Graduate Teaching Assistant, MATH 112 (College Algebra), Sections 101, 102, 201, 202, 401 & 402 – Spring 2024
- Graduate Teaching Assistant, MATH 125 (Calculus I), Section 001 – Fall 2024
- Graduate Teaching Assistant, MATH 112 (College Algebra), Sections 103 & 203 – Spring 2025
- **Teaching Mentors & Advisors:** Mitchell Wilson, Tina Deemer, Catherine Yslas, Oussama Ben Said, Tynan Lazarus, and Prof. David Glickenstein

San Francisco State University

- Graduate Teaching Assistant, Calculus – Spring 2022

- Grader, MATH 227 [05] (Calculus II)
- Instructor, MATH 226 [38] (Calculus I) – Fourth-hour component of MATH 226 [37]
- Instructor, MATH 227 [06] (Calculus II) – Fourth-hour component of MATH 227 [05]
- Instructor, MATH 227 [36] (Calculus II) – Fourth-hour component of MATH 227 [35]
- **Advisors:** Prof. Kim Seashore, Prof. Shandy Hauk, and Prof. Eric Hsu

Other Academic Experience

San Francisco State University

- Graduate Teaching Assistant, Pre-Calculus – Fall 2019
- Advisor: Prof. Kim Seashore

University of San Francisco

- San Francisco Math Circle – Fall 2016
- Advisor: Prof. Paul Zeitz

National Dong Hwa University

- Undergraduate Research Assistant – Spring 2010 Hired and advised by Prof. Cheng-Pang Liu
- Tutor of Calculus and General Physics – August 2008 to December 2009 Hired by the NDHU Department of Physics

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Awards and Honors

- **Nominated for MSRI Summer Graduate School on Metric Geometry and Geometric Analysis**
  University of Oxford (UK), Fall 2021
- **Dean's Honor Roll**
  University of San Francisco, Spring 2018
- **Mathematics Advanced Study Scholarship and Internal Scholarship (MASS Program)**
  The Pennsylvania State University–University Park, Fall 2017
  (Covered tuition and fees)
- **Dean's Honor Roll**
  University of San Francisco, Spring 2015, Fall 2016, and Spring 2017
- **Pi Mu Epsilon Honor Society**
  University of San Francisco
- **Admitted to the Summer School on Symmetry in Mathematics and Physics**
  National Taiwan University, Summer 2012
- **Admitted to Prof. Anthony Zee's Quantum Field Theory Course**
  Institute of Physics, Academia Sinica, February 2012
- **Admitted to the 1st LeCosPA Symposium: Towards Ultimate Understanding of the Universe**

National Taiwan University, February 2012
[Link]

- **Admitted to the 2nd International Workshop on Dark Matter, Dark Energy, and Matter-Antimatter Asymmetry**
  National Tsing Hua University, Winter 2010
  [Link]
- **Admitted to the Summer School for Theoretical Physics**
  National Tsing Hua University, Summer 2009
- **President's List**
  National Dong Hwa University, March 2008, November 2008, March 2009, March 2010

Certificates

- **Protecting God's Children Online Awareness 4.0**
  Order of Malta, Western Association – March 7, 2025
- **Safety Preparedness Training**
  The University of Arizona, Employee Development, Growth, and Engagement
  December 8, 2023
- **Information Security Awareness Certification**
  The University of Arizona, Employee Development, Growth, and Engagement
  August 27, 2023
- **MASS Program Completion**
  Completed all requirements for the 2017 Mathematics Advanced Study Semesters program at The Pennsylvania State University
- **Recognition of Service Award**
  ACM Special Interest Group on Management of Data (SIGMOD) – 2016
- **Tackling the Challenges of Big Data**
  Online program developed by the faculty of the MIT Computer Science and Artificial Intelligence Laboratory
  February 3 – March 17, 2015

Service to the Community

Early Service Initiatives (1998 – 2003).

- 1998 – 2003 – Altar Server (Mass Assistant) at Fu Jen Catholic University, assisting at liturgical celebrations attended by university professors and students.
- 2000 – First engagement with service learning at Boston College, accompanying my father, a seed professor for Fu Jen Catholic University's Service Learning Program. Taught elementary arithmetic to children from minority communities in Boston.

High School Service (2000 – 2006).

- 2000 – 2006 – Served a minimum of 15 hours per year at local hospitals in Taipei through St. Ignatius High School, assisting:
  - Elderly patients, feeding them spoon by spoon.
  - Children with cerebral palsy and other disabilities, ensuring they received meals with care and dignity.

University and Adult Service (2017 – Present).

- 2017 (Spring) – Teaching Assistant for Mathematics Enrichment Program at the University of San Francisco (USF), providing academic support and instructional assistance to students.
- 2023 (October 16 – Present) – Volunteer at the University of Arizona Tucson Math Circle, primarily maintaining the website to support outreach and mathematical education.
- 2025 (February 21) – Began volunteering at Poverello House (Tucson, AZ), assisting the homeless community (4-hour shift).
- 2025 (March 29, scheduled) – Continuing volunteer service at Poverello House (Tucson).
- 2025 (May 18, upcoming) – Volunteering at a Blood Drive (9 AM – 12 PM) organized by the Knights of Columbus at St. Thomas the Apostle Parish (Tucson, AZ).

Future Service Commitment.

- Following my graduation from the University of Arizona (Spring 2025), I plan to increase my volunteer engagement with the Order of Malta, Poverello House, the Knights of Columbus, Catholic Action in San Diego, and other service initiatives.

Faith Formation and Study.

- Regular recipient and self-study of Opus Dei's publications, integrating its spiritual insights into daily life, professional service, and ethical decision-making.
- Active engagement with Catholic Action in San Diego, remotely supporting its evangelization efforts and operations through prayer and spiritual solidarity.
- Ongoing study of books published by the Order of Malta and its members, including the Catholic Bible, prayer books, and *Minutes with the Catechism*, deepening my understanding of faith and service.

Skills

- **Problem Solving and Adaptability:** Demonstrated ability to learn new skills quickly.
- **Programming Languages:** C/C++, Python, R, Java, Lisp, Shell Script, Sed, Awk, LaTeX, Mathematica
- **Libraries and Packages:** PyTorch, Lightning, NumPy, Pandas, scikit-learn, Matplotlib, Ogre3D
- Designing algorithms to generate examples for theoretical research in mathematics, physics, statistics, and computer science